with probability at least $1-O(\epsilon)$. Using the union bound $M_V^T = M_V q_{T(\ell)}(1 \pm M_V^{-1/2} q_{T(\ell)}^{-1/2}\sqrt{\log(L\epsilon^{-1})})$, the matrix Hoeffding's inequality, standard Hoeffding's inequality, and $\|\mathbf{1}_{1:N}\boldsymbol{x}_t^{(i_T)\top}\|_2 \lesssim 1$, we obtain

$$
\begin{aligned}
&\left\| \frac{1}{M_V} \sum_{T=5}^{L-1} \sum_{i_T=1}^{M_V^T} \left(\frac{1}{TN} \sum_{t=1}^{T} \mathbf{1}_{1:N} \boldsymbol{x}_t^{(i_T)\top} \right) - \mathbb{E}\left(\frac{1}{TN} \sum_{t=1}^{T} \mathbf{1}_{1:N} \boldsymbol{x}_t^{(i_T)\top} \right) \right\|_2 \\
\lesssim &\left\| \sum_{T=5}^{L-1} \frac{M_V^T}{M_V} \frac{1}{M_V^T} \sum_{i_T=1}^{M_V^T} \frac{1}{TN} \boldsymbol{A}^{(i_T)} \right\|_2 + \left\| \sum_{T=5}^{L-1} \left(\frac{M_V^T}{M_V} - q_{T(\ell)} \right) \mathbb{E}\left(\frac{1}{TN} \sum_{t=1}^{T} \mathbf{1}_{1:N} \boldsymbol{x}_t^{(i_T)\top} \mid T \right) \right\|_2 \\
\lesssim &\sum_{T=5}^{L-1} \underbrace{\frac{M_V^T}{M_V}}_{\simeq \mathrm{Prob(T)}} \underbrace{\left\| \frac{1}{M_V^T} \sum_{i_T=1}^{M_V^T} \frac{1}{TN} \boldsymbol{A}^{(i_T)} \right\|_2}_{\sqrt{\frac{\log\left(L(N+L)\epsilon^{-1}\right)}{NM_V \mathrm{Prob(T)}}}} + \sum_{T=5}^{L-1} \underbrace{\left| \frac{M_V^T}{M_V} - q_{T(\ell)} \right|}_{\lesssim \frac{\sqrt{q_{T(\ell)} \log\left(L\epsilon^{-1}\right)}}{\sqrt{M_V}}} \underbrace{\left\| \mathbb{E}\left(\frac{1}{TN} \sum_{t=1}^{T} \mathbf{1}_{1:N} \boldsymbol{x}_t^{(i_T)\top} \mid T \right) \right\|_2}_{N^{-1/2}} \\
\lesssim &\frac{\sum_\ell \sqrt{q_\ell}}{\sqrt{NM_V}} \sqrt{\log\left(L(N+L)\epsilon^{-1}\right)}.
\end{aligned}
$$

Let $\bar{\boldsymbol{B}}_{i_{T,k}} = \sum_{t=1}^{T} \boldsymbol{e}_k \boldsymbol{x}_t^{(i_{T,k})\top}$. Using $M_V^{T,k} \simeq M_V^T (N - N_{\mathrm{trg}})^{-1}(1 \pm (M_V^T)^{-1/2} N^{1/2} \sqrt{\log(NL\epsilon^{-1})})$, $M_V^T = M_V q_{T(\ell)}(1 \pm M_V^{-1/2} q_{T(\ell)}^{-1/2}\sqrt{\log(L\epsilon^{-1})})$, the matrix Hoeffding's inequality, standard Hoeffding's inequality, and $\|T^{-1}\bar{\boldsymbol{B}}_{i_{T,k}}\|_2 \lesssim 1$, we obtain

$$
\begin{aligned}
&\Bigg\| \sum_{T=5}^{L-1} \frac{M_V^T}{M_V} \sum_{k=N_{\mathrm{trg}}+1}^{N} \frac{M_V^{T,k}}{M_V^T} \frac{1}{M_V^{T,k}} \sum_{i_{T,k}=1}^{M_V^{T,k}} \left(\frac{1}{T} \sum_{t=1}^{T} \boldsymbol{e}_k \boldsymbol{x}_t^{(i_{T,k})\top} \right) \\
&- \sum_{k=N_{\mathrm{trg}}+1}^{N} \frac{1}{N - N_{\mathrm{trg}}} \mathbb{E}_T \left[\mathbb{E}\left[\left(\frac{1}{T} \sum_{t=1}^{T} \boldsymbol{e}_k \boldsymbol{x}_t^{(i_{T,k})\top} \right) \mid y = k \right] \right] \Bigg\|_2 \\
\lesssim &\left\| \sum_{T=5}^{L-1} \frac{M_V^T}{M_V} \sum_{k=N_{\mathrm{trg}}+1}^{N} \frac{M_V^{T,k}}{M_V^T} \frac{1}{M_V^{T,k}} \sum_{i_{T,k}=1}^{M_V^{T,k}} T^{-1} \boldsymbol{B}^{(i_{T,k})} \right\|_2 \\
&+ \left\| \sum_{T=5}^{L-1} \frac{M_V^T}{M_V} \sum_{k=N_{\mathrm{trg}}+1}^{N} \left(\frac{M_V^{T,k}}{M_V^T} - \frac{1}{N - N_{\mathrm{trg}}} \right) \frac{1}{T} \mathbb{E}\left[\bar{\boldsymbol{B}}^{(i_{T,k})} \mid y = k, T \right]) \right\|_2 \\
&+ \left\| \sum_{T=5}^{L-1} \left(\frac{M_V^T}{M_V} - q_{T(\ell)} \right) \sum_{k=N_{\mathrm{trg}}+1}^{N} \frac{1}{N - N_{\mathrm{trg}}} \mathbb{E}[\bar{\boldsymbol{B}}_{i_{T,k}} | k, T] \right\|_2 \\
\lesssim &\sum_{T=5}^{L-1} \underbrace{\frac{M_V^T}{M_V}}_{q_{T(\ell)}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \underbrace{\frac{M_V^{T,k}}{M_V^T}}_{N^{-1}} \underbrace{\left\| \frac{1}{M_V^{T,k}} \sum_{i_{T,k}=1}^{M_V^{T,k}} T^{-1} \boldsymbol{B}^{(i_{T,k})} \right\|_2}_{\sqrt{\frac{N \log\left(NL(N+L)\epsilon^{-1}\right)}{M_V \mathrm{Prob(T)}}}} \\
&+ \sum_{T=5}^{L-1} \frac{M_V^T}{M_V} \sum_{k=N_{\mathrm{trg}}+1}^{N} \underbrace{\left| \frac{M_V^{T,k}}{M_V^T} - \frac{1}{N - N_{\mathrm{trg}}} \right|}_{\sqrt{\frac{\log\left(NL\epsilon^{-1}\right)}{NM_V q_{T(\ell)}}}} \underbrace{\left\| \frac{1}{T} \mathbb{E}\left[\bar{\boldsymbol{B}}^{(i_{T,k})} \mid y = k, T \right]) \right\|_2}_{\lesssim 1} \\
&+ \sum_{T=5}^{L-1} \left| \frac{M_V^T}{M_V} - q_{T(\ell)} \right| \left\| \sum_{k=N_{\mathrm{trg}}+1}^{N} \frac{1}{N - N_{\mathrm{trg}}} \mathbb{E}[\bar{\boldsymbol{B}}_{i_{T,k}} | k, T] \right\|_2
\end{aligned}
$$

$$\lesssim \left(\sum_{\ell} \sqrt{q_\ell}\right) \sqrt{\frac{N \log\left(NL(N+L)\epsilon^{-1}\right)}{M_V}}.$$

□

Note that if $M_V \gtrsim \operatorname{poly}\log N \cdot \frac{N^5}{N_{\mathrm{trg}}^2}\left(\frac{\sum_{\ell\in\mathcal{S}}\sqrt{q_\ell}}{\sum_{\ell\in\mathcal{S}} q_\ell \ell^{-1}}\right)^2$, then we obtain $\|\bar{\boldsymbol{W}}_V^* - \boldsymbol{W}_V^*\|_2 \lesssim \tilde{\eta}_V \mathbb{E}[T^{-1}] N_{\mathrm{trg}} N^{-1}$, where $\bar{\boldsymbol{W}}_V^*$ and $\boldsymbol{W}_V^*$ are $\boldsymbol{W}_V$ after one GD step with infinite and finite sample size, respectively. The following corollary is obtained by combining Lemmas 9 and 17.

Corollary 18. *If we set $\eta_V \lesssim 1$ and $M_V \gtrsim \operatorname{poly}\log N \cdot \frac{N^5}{N_{\mathrm{trg}}^2}\left(\frac{\sum_{\ell\in\mathcal{S}}\sqrt{q_\ell}}{\sum_{\ell\in\mathcal{S}} q_\ell \ell^{-1}}\right)^2$, then it holds that $|1/N - p(k|z)| = O(\mathbb{E}[T^{-1}]/N^2)$ for any k and z, where $p(k|z)$ is the transformer output regarding token k after pretraining $\boldsymbol{W}_V$.*

C.2 Key-Query matrix

Now let us consider the KQ-matrix with finite samples

$$\boldsymbol{W}_{KQ} = -\eta_{KQ} \sum_{T=5}^{L-1} \sum_{k=N_{\mathrm{trg}}+1}^{N} \left(\frac{M_{KQ}^T}{M_{KQ}} \hat{\boldsymbol{C}}(T,k) - \frac{M_{KQ}^T}{M_{KQ}} \frac{M_{KQ}^{T,k}}{M_{KQ}^T} \hat{\boldsymbol{D}}(T,k) \right) + \eta_{KQ} \tilde{\eta_V} \mathbb{E}[T^{-1}] O_\infty(N_{\mathrm{trg}}/N)$$

where

$$\begin{aligned}\hat{\boldsymbol{C}}(T,k) \coloneqq & \sum_{i_T} \frac{1}{M_{KQ}^T} \frac{1}{N - N_{\mathrm{trg}}} \left(\left(\frac{1}{T} - \frac{1}{T^2} \right) \sum_t \boldsymbol{x}_t^{(i_T)} \boldsymbol{x}_t^{(i_T)\top} \boldsymbol{W}_V^\top \boldsymbol{e}_k \boldsymbol{x}_T^{(i_T)\top} \right) \\ & - \sum_{i_T} \frac{1}{M_{KQ}^T} \frac{1}{N - N_{\mathrm{trg}}} \left(\frac{1}{T^2} \sum_{t \neq t'} \boldsymbol{x}_t^{(i_T)} \boldsymbol{x}_{t'}^{(i_T)\top} \boldsymbol{W}_V^\top \boldsymbol{e}_k \boldsymbol{x}_T^{(i_T)\top} \right)\end{aligned}$$

and

$$\begin{aligned}\hat{\boldsymbol{D}}(T,k) \coloneqq & \sum_{i_{T,k}} \frac{1}{M_{KQ}^{T,k}} \left(\left(\frac{1}{T} - \frac{1}{T^2} \right) \sum_t \boldsymbol{x}_t^{(i_{T,k})} \boldsymbol{x}_t^{(i_{T,k})\top} \boldsymbol{W}_V^\top \boldsymbol{e}_k \boldsymbol{x}_T^{(i_{T,k})\top} \right) \\ & - \sum_{i_{T,k}} \frac{1}{M_{KQ}^{T,k}} \left(\frac{1}{T^2} \sum_{t \neq t'} \boldsymbol{x}_t^{(i_{T,k})} \boldsymbol{x}_{t'}^{(i_{T,k})\top} \boldsymbol{W}_V^\top \boldsymbol{e}_k \boldsymbol{x}_T^{(i_{T,k})\top} \right)\end{aligned}$$

where

$$\boldsymbol{W}_V^\top \boldsymbol{e}_k = \tilde{\eta_V} \mathbb{E}[T^{-1}] \left[\begin{array}{c} 0 \\ \vdots \\ 0 \\ \hline \boldsymbol{0}_{k-1} \\ 1 \\ \boldsymbol{0}_{N-k} \\ \hline \boldsymbol{0}_{k-1} \\ 1 \\ \boldsymbol{0}_{N-k} \end{array} \right] + o(\tilde{\eta_V} \mathbb{E}[T^{-1}]) \cdot \boldsymbol{1}_{L+2N}.$$

Lemma 19. *Let $\boldsymbol{B}_{t,t',T,k} = \boldsymbol{x}_t (\boldsymbol{x}_{t'}^\top \boldsymbol{W}_V \boldsymbol{e}_k) \boldsymbol{x}_T^\top$. Then, we have*

$$\boldsymbol{B}_{t,t',T,k} = \left((\mathbb{1}[z_{t'} = k] + \mathbb{1}[z_{t'-1} = k]) \tilde{\eta_V} \mathbb{E}[T^{-1}] + o(\tilde{\eta_V} \mathbb{E}[T^{-1}]) \right) \boldsymbol{x}_t \boldsymbol{x}_T^\top.$$

Therefore,

$$\begin{bmatrix} \boldsymbol{B}_{t,t',T,k}\boldsymbol{B}_{t,t',T,k}^\top & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{B}_{t,t',k}^\top \boldsymbol{B}_{t,t',k} \end{bmatrix} \preceq \boldsymbol{\Sigma}$$

where $\lambda_i(\boldsymbol{\Sigma}) = O(\tilde{\eta_V}\mathbb{E}[T^{-1}])^2$ *for* $i = 1, 2$ *and* $\lambda_i(\boldsymbol{\Sigma}) = 0$ *for* $i > 3$.

Proof. The proof is straightforward. Note that $\boldsymbol{x}_{t'}^\top \mathbf{1}_{L+2N} = O(1)$ for all t'. □

Lemma 20. *Let* $\bar{\boldsymbol{W}}_{KQ}^*$ *be* $\boldsymbol{W}_{KQ}$ *after one gradient descent step with finite sample size (Algorithm 1) and* $\boldsymbol{W}_{KQ}^*$ *be the counterpart for infinite sample size. Let* $\left\{\left\{\boldsymbol{x}_t^{(i)}\right\}_{t=1}^{T_i}\right\}_{i=1}^{M_V}$ *be i.i.d. sample sequences,* $\left\{\left\{\boldsymbol{x}_t^{(i_T)}\right\}_{t=1}^{T}\right\}_{i_t=1}^{M_V^T}$ *be conditionally i.i.d. sub-samples with fixed* T*, and* $\left\{\left\{\boldsymbol{x}_t^{(i_{T,k})}\right\}_t\right\}_{i_{T,k}=1}^{M_V^{T,k}}$ *be conditionally i.i.d. sub-samples with fixed* T *and* $z_{T+1} = k$*. With probability at least* $1 - O(\epsilon)$*,*

$$\lambda_{\max}\left(\bar{\boldsymbol{W}}_{KQ}^* - \boldsymbol{W}_{KQ}^*\right) \lesssim \eta_{KQ}\tilde{\eta_V}\mathbb{E}[T^{-1}]\left(\sum_l \sqrt{q_\ell}\right)\sqrt{\frac{N\log\left(LN(L+N)\epsilon^{-1}\right)}{M_{KQ}}} + \eta_{KQ}\tilde{\eta_V}\mathbb{E}[T^{-1}]O_\infty(N_{\mathrm{trg}}/N).$$

Proof. We have

$$\left\|\begin{bmatrix} \hat{\boldsymbol{C}}(T,k)\hat{\boldsymbol{C}}(T,k)^\top & \boldsymbol{O} \\ \boldsymbol{O} & \hat{\boldsymbol{C}}(T,k)^\top\hat{\boldsymbol{C}}(T,k) \end{bmatrix}\right\|_2 \lesssim (\tilde{\eta_V}\mathbb{E}[T^{-1}])^2$$

and

$$\left\|\begin{bmatrix} \hat{\boldsymbol{D}}(T,k)\hat{\boldsymbol{D}}(T,k)^\top & \boldsymbol{O} \\ \boldsymbol{O} & \hat{\boldsymbol{D}}(T,k)^\top\hat{\boldsymbol{D}}(T,k) \end{bmatrix}\right\|_2 \lesssim (\tilde{\eta_V}\mathbb{E}[T^{-1}])^2.$$

by Lemma 19. Then we obtain the error bound as

$$\begin{aligned}
&\frac{1}{\eta_{KQ}}\left\|\boldsymbol{W}_{KQ} - \boldsymbol{W}_{KQ}^*\right\|_2 \\
\lesssim&\left\|\sum_{T=5}^{L-1}\frac{M_{KQ}^T}{M_{KQ}}\left(\hat{\boldsymbol{C}}(T,k) - \mathbb{E}[\hat{\boldsymbol{C}}(T,k)|T]\right)\right\|_2 + \left\|\sum_{T=5}^{L-1}\left(\frac{M_{KQ}^T}{M_{KQ}} - q_{T(\ell)}\right)\mathbb{E}[\hat{\boldsymbol{C}}(T,k)|T]\right\|_2 \\
&+\left\|\sum_{T=5}^{L-1}\frac{M_{KQ}^T}{M_{KQ}}\sum_{k=N_{\mathrm{trg}}+1}^{N}\frac{M_{KQ}^{T,k}}{M_{KQ}^T}\left(\hat{\boldsymbol{D}}(T,k) - \mathbb{E}[\hat{\boldsymbol{D}}(T,k)|T,k]\right)\right\|_2 \\
&+\left\|\sum_{T=5}^{L-1}\frac{M_{KQ}^T}{M_{KQ}}\sum_{k=N_{\mathrm{trg}}+1}^{N}\left(\frac{M_{KQ}^{T,k}}{M_{KQ}^T} - \frac{1}{N-N_{\mathrm{trg}}}\right)\mathbb{E}[\hat{\boldsymbol{D}}(T,k)|T,k]\right\|_2 \\
&+\left\|\sum_{T=5}^{L-1}\left(\frac{M_{KQ}^T}{M_{KQ}} - q_{T(\ell)}\right)\mathbb{E}[\hat{\boldsymbol{D}}(T,k)|T]\right\|_2 + \eta_{KQ}\tilde{\eta_V}\mathbb{E}[T^{-1}]O_\infty(N_{\mathrm{trg}}/N) \\
\lesssim&\tilde{\eta_V}\mathbb{E}[T^{-1}]\left(\sum_\ell \sqrt{q_\ell}\right)\sqrt{\frac{N\log\left((L+N)\epsilon^{-1}\right)}{M_{KQ}}} + \tilde{\eta_V}\mathbb{E}[T^{-1}]O_\infty(N_{\mathrm{trg}}/N)
\end{aligned}$$

in the same way as bounding $\boldsymbol{W}_V$. We used $M_{KQ}^T \simeq q_{\ell(T)}M_{KQ}$, $M_{KQ}^{T,k} \simeq N^{-1}M_{KQ}^T$, (matrix-) Hoeffding's inequalities, and $\|\hat{\boldsymbol{C}}(T,k)\|_2, \|\hat{\boldsymbol{D}}(T,k)\|_2 \lesssim \tilde{\eta}_V\mathbb{E}[T^{-1}]$ by Lemma 19. □

Based on these finite sample analyses, Theorem 6 can be obtained similarly to Theorem 5.

Proof. [Proof of Theorem 6] Note that, the approximation in Section B.2.1 still applies, if the finite-sample error with respect to $\boldsymbol{W}^{KQ}$ is falling into $\tilde{\eta}O(N_{\mathrm{trg}}/N)$. From Lemma 20, it suffices to set $M_{KQ} \gtrsim \mathrm{poly}\log N \cdot$